

MUHAMMAD HASSAN

AI/ML Engineer | Software Development | Automation Engineering

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PROFESSIONAL SUMMARY

Final-year Computer Science student at Lahore Garrison University building production-style AI/ML systems and full-stack applications — from training ensemble classifiers and writing a from-scratch compiler to deploying Flask APIs and automation pipelines. Comfortable working across the stack: React frontends, n8n workflows, and Claude API integrations. Seeking a software development internship where I can contribute practically and grow within an engineering team.

EDUCATION

Bachelor of Science in Computer Science (BSCS)

Expected 2027

Lahore Garrison University | Currently: Final Year

- Relevant Coursework: Machine Learning, Parallel & Distributed Computing, Compiler Construction, Software Quality Assurance, Cloud Computing, Database Systems

PROJECTS

Smart Healthcare Prediction System

Python · Flask · React · Scikit-learn · Pandas · NumPy

- Benchmarked 4 classifier types — Random Forest, Gradient Boosting, SVM, and Logistic Regression — on a class-imbalanced clinical dataset (mirroring UCI Diabetes 130-US Hospitals data), then combined them in a Voting Classifier ensemble.
- Applied SMOTE for class rebalancing and StandardScaler for feature normalization, then served predictions through a Flask REST API connected to a React frontend for real-time inference via web interface.
- Voting Classifier achieved the best test F1 of 0.60 across 3 classes (Non-Diabetic, Pre-Diabetic, Diabetic); results documented in a formal academic report covering model architecture, evaluation metrics, and design decisions.

GitHub: github.com/muhammadhassan-ai/machine-learning-project

SentinelX — Mini Compiler Assistant

Python · Flask · PLY (Lex/Yacc)

- Implemented a full compiler pipeline in Python covering all 5 stages: lexical analysis, parsing (context-free grammar), semantic analysis, intermediate representation (IR) generation with 4 optimization passes, and a tree-walking interpreter.
- Exposed the compiler pipeline through a Flask web interface, allowing users to input source code and inspect output at each compilation stage — lexer tokens, AST, IR, and execution result.
- Supports a statically-typed mini language with integer, float, string, and boolean types; handles arithmetic, comparison, and logical operators, if-else branching, and while loops with runtime guards; detects and reports 7 distinct error types across compilation stages.

GitHub: github.com/muhammadhassan-ai/SentinelX

Roman Pizza Website

React · Node.js · CSS · Selenium

- Built a full-stack pizza ordering website with a dynamic React frontend showcasing menu items, an interactive cart, and order placement flow — deployed and publicly available.
- Implemented a clean, mobile-responsive UI with component-based architecture, demonstrating practical front-end engineering skills beyond academic projects.
- Built and ran a Selenium-based automated test suite covering the ordering flow as the formal Software Quality Assurance (SQA) coursework deliverable for this project.

GitHub: github.com/muhammadhassan-ai/Roman-pizza-website

Instagram DM Auto-Reply Agent

Node.js · ManyChat API · Claude API

- Automated Instagram DM handling for a business client by combining rule-based routing with Claude API-generated replies, eliminating manual response effort and enabling around-the-clock coverage.
- Integrated ManyChat API with a Node.js backend to route and respond to incoming messages at scale without human intervention.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, HTML, CSS

Frameworks & Libraries: React, Node.js, Flask, Scikit-learn, Pandas, NumPy, Matplotlib

AI & Automation: Claude API (Anthropic), n8n, ManyChat, ML model deployment

Databases & Tools: MySQL, Git, GitHub, VS Code

Networking: Cisco Packet Tracer

Concepts: Machine Learning, REST APIs, OOP, Data Analysis, Cloud Computing, Compiler Design

CERTIFICATIONS

Elements of AI — University of Helsinki & MinnaLearn

June 2026

Google AI Agents — Google Skills

March 2025